

RAJESH S

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ABOUT ME

Passionate backend developer with hands-on experience in building real-world web and AI-powered applications. Strong interest in backend architecture, API development, and cloud deployment. Enjoys solving practical problems through clean, scalable, and efficient systems.

SKILLS

Languages: Python · Java · C · SQL · PL/SQL

Frameworks: FastAPI · Flask · React · REST API · API Design · Microservices · YOLOv8 · OpenCV · LangChain

AI / ML: GPT-4o · FAISS · PyTorch · MTCNN · XGBoost · SHAP · SMOTE · Scikit-learn · Pandas · NumPy

DevOps & Tools: Docker · AWS EC2 · CI/CD · Git · Postman · MySQL · MongoDB · Streamlit · Cloud Deployment

EXPERIENCE

AI Chatbot Developer

Aug 2025 – Sep 2025

Loyola Institute of Business Administration (LIBA) · Python, FastAPI, GPT-4o, FAISS, AWS EC2

- Architected and deployed a production RAG-based FAQ chatbot on liba.edu, reducing manual query handling by ~30% through API design and cloud deployment on AWS EC2.
- Integrated GPT-4o for NLU and FAISS vector search for context retrieval; improved chatbot response accuracy by ~20% via prompt tuning and context ranking.
- Designed microservices-style query processing pipeline with fallback handling; conducted unit testing to ensure robust, production-ready responses.

Backend Developer Intern

Nov 2025 – Jan 2026

Upstride · Python, FastAPI, Scikit-learn, Pandas, React, Docker

- Designed and built 10+ RESTful APIs with FastAPI following API design best practices, powering backend ML workflows with improved reliability.
- Set up CI/CD pipelines for automated testing and deployment; conducted unit testing achieving 90%+ test coverage on critical modules.
- Performed feature engineering and model evaluation, improving ML model accuracy by 15%; documented pipelines and API endpoints for team maintainability.

PROJECTS

Enterprise Fraud Risk Intelligence Platform · XGBoost, SHAP, SMOTE, Streamlit, Scikit-learn

- Built end-to-end fraud detection system generating a 0–100% Risk Score with per-transaction SHAP explainability, cutting analyst review time by 40%.
- Tuned XGBoost with RandomizedSearchCV and early stopping — achieved ROC-AUC 0.973 and PR-AUC 0.819 on imbalanced data via SMOTE.
- Implemented drift monitoring (PSI + KS tests) and a 5-page Streamlit dashboard; deployed on cloud for real-time analyst access.
- Engineered modular feature pipeline covering transaction velocity, behavioural patterns, and account-age signals for robust fraud detection.

AI Smart CCTV & Hazard Detection · YOLOv8, OpenCV, FastAPI, React, MongoDB

- Engineered CV pipeline converting smartphones into IP cameras with zero-latency YOLOv8 hazard detection, supporting 5+ concurrent streams.
- Designed microservices-based FastAPI backend with async processing; stored incident metadata in MongoDB and triggered instant Telegram alerts.
- Built live React dashboard for concurrent stream viewing and incident history; reduced hazard response time by ~50% vs. manual monitoring.
- Implemented CI/CD pipeline for automated cloud deployment, ensuring zero-downtime updates and consistent delivery across environments.

AI Voice Assistant – Virtual HR Interview System · Python, FastAPI, Pydantic, Uvicorn, Web Speech API

- Designed voice-interactive interview simulator with adaptive question flow across 3 difficulty levels (Beginner, Intermediate, Advanced).
- Scored answers using STAR method, keyword matching, and response clarity — generating real-time performance reports with structured feedback.
- Exposed RESTful API endpoints via FastAPI + Uvicorn; Web Speech API enables fully hands-free speech recognition and TTS with <200ms latency.
- Implemented session management tracking question history and preventing repetition, improving candidate experience across 50+ test sessions.

Face Detection & Clustering System · PyTorch, MTCNN, InceptionResnetV1, DBSCAN, t-SNE/UMAP

- Automated pipeline detecting faces with MTCNN and extracting 512-dim embeddings via InceptionResnetV1, processing 1,000+ images per run.
- Applied DBSCAN, Hierarchical, and Chinese Whispers clustering; evaluated with silhouette, NMI & ARI metrics — achieving 90%+ grouping accuracy.
- Generated t-SNE / UMAP visualisations and summary dashboard; auto-organised results into identity folders, reducing manual sorting by 100%.
- Handled diverse real-world datasets with varied lighting, angles, and occlusions — ensuring robust detection and accurate identity grouping.

EDUCATION

B.Tech – Artificial Intelligence & Data Science

Jun 2024 – May 2028

Loyola College of Engineering and Technology | Chennai, India

EXTRACURRICULAR

- AI TestGen presented at TN-IMPACT @ KIT (Mar 2026) — achieved 60–80% faster test preparation vs. manual methods.
- Hackathons: Google Gemini · Smart India Hackathon (SIH) · SRM Kattankulathur (IoT) · 30-hour Hackathon.
- NSS Volunteer: orphanage visits, village & school outreach, and community awareness campaigns.